# 2026-04-26-jackett-autoconfig-clean-rebuild-design

# Jackett Auto-Configuration + Clean-Slate Rebuild — Design

**Revision:** 1 **Last modified:** 2026-06-06T00:00:00Z

**Date:** 2026-04-26 **Author:** brainstorming session (Claude + project owner) **Status:** approved, ready for implementation plan **Implementation skill next:** `superpowers:writing-plans`

## 1. Problem & Intent

The project owner wants three things in one pass:

1. **Clean-slate rebuild** of the Python-profile container stack so the system boots from a known-zero state.
2. **Full Jackett indexer integration** — every credentialed tracker we have an `.env` triple for should be auto-configured in Jackett at proxy startup, with no manual Jackett UI clicks. New trackers added to `.env` later should pick themselves up automatically.
3. **Comprehensive tests + challenges** at every CONSTITUTION-mandated layer, plus a parity audit so the eventual Go-only flip doesn't silently drop features.

Closing instruction: commit and push to all upstreams as work progresses; hand off when the stack is verified and ready for manual testing.

# 2. Decisions Locked During Brainstorming

| # | Question | Choice | Implication |
|---|---|---|---|
| Q1 | What does “all Jackett trackers” mean? | (c) Auto-enable + credential injection from `.env` | We write code that configures Jackett on the user’s behalf for indexers we have creds for. We do **not** auto-enable Jackett’s 600+ public indexers blindly. |
| Q2 | Which credentialed indexers? | (b) Generic env discovery + fuzzy match + override | RuTracker, Kinozal, NNMClub, IPTorrents today; any future `<NAME>_USERNAME/_PASSWORD/_COOKIES` triple gets picked up automatically; `JACKETT_INDEXER_MAP` resolves ambiguous matches. |
| Q3 | What gets wiped on “clean slate”? | (b) Containers + images + `./config/jackett` | Keep `./config/qBittorrent`, `./tmp`, `/mnt/DATA`. Forces auto-config to bootstrap from zero, which is the test we want. |
| Q4/Q5 | Python or Go for the rebuild? | (d) Python now, parity audit doc, Go port deferred | Implementation runs against Python proxy. A `docs/migration/PARITY_GAPS.md` deliverable catalogs what’s still unported in Go so the eventual flip loses nothing. |
| Q6 | Test coverage floor? | (a) Full CONSTITUTION compliance | All 7 layers: unit, integration, E2E, security/penetration, benchmark, automation (mapped to `tests/` |

| # | Question | Choice | Implication |
|---|---|---|---|
| | | | `contract/` — see §5), challenge. |
| Q7 | When does push happen? | (c) Per-layer commit + push | After each test layer goes green, commit + push to all 3 remotes. |
| Approach | Module shape? | (1) Module-in-`merge_service`, idempotent, startup-time | Lowest topology change, highest testability, easy to lift-and-shift to Go later. |

# 3. Architecture

## 3.1 Module

**New file:** `download-proxy/src/merge_service/jackett_autoconfig.py`

**Public API (single function):**

```python
async def autoconfigure_jackett(
    jackett_url: str,
    api_key: str,
    env: Mapping[str, str],
    indexer_map_override: dict[str, str] | None = None,
    timeout: float = 10.0,
) -> AutoconfigResult: ...
```

**`AutoconfigResult`** is a Pydantic model with: - `ran_at: datetime` - `discovered_credentials: list[str]` — tracker names found in env (NEVER includes the values) - `matched_indexers: dict[str, str]` — env name → Jackett indexer ID - `configured_now: list[str]` — indexer IDs configured this run - `already_present: list[str]` — indexer IDs Jackett already had configured (idempotency hits) - `skipped_no_match: list[str]` — env names where fuzzy match failed - `skipped_ambiguous: list[dict]` — env name + candidate IDs where ≥2 indexers matched - `errors: list[str]` — structured error codes (e.g., `jackett_unreachable`, `jackett_auth_failed`, `catalog_parse_failed`, `indexer_config_failed:<id>`)

The function never raises out of its boundary; all failures land in `errors`.

## 3.2 Call site

`download-proxy/src/main.py` — after `JACKETT_API_KEY` is verified non-empty (the existing startup already waits for Jackett's key extraction), and before the FastAPI app starts serving traffic. Failure is logged WARNING; **boot continues**. Auto-config is best-effort.

## 3.3 Read endpoint

`GET /merge/jackett/autoconfig/last` — returns the cached `AutoconfigResult` from the most recent run. Read-only. Lives in `download-proxy/src/api/routes.py`. Response shape:

```
{
  "ran_at": "2026-04-26T14:23:11Z",
  "discovered": ["RUTRACKER", "KINOZAL", "NNMCLUB", "IPTORRENTS"],
  "configured_now": ["rutracker", "kinozalbiz"],
  "already_present": ["nnmclub"],
  "skipped_no_match": ["IPTORRENTS"],
  "skipped_ambiguous": [],
  "errors": []
}
```

Credentials NEVER appear in the response, the cached object's `__repr__`, or any log line at any level.

## 3.4 Discovery algorithm (3 passes)

1. **Env scan** — regex `^([A-Z][A-Z0-9_]+?)_(USERNAME|PASSWORD|COOKIES)$` over `os.environ`. Group by tracker name. A "credential bundle" is valid when the bundle contains either `(USERNAME && PASSWORD)` or `COOKIES`. Names matching the **denylist** (default `QBITTORRENT,JACKETT,WEBUI,PROXY,MERGE,BRIDGE`, overridable via `JACKETT_AUTOCONFIG_EXCLUDE` comma-separated env var) are dropped before pass 2 — these are project-internal credentials (qBit auth, Jackett's own key, etc.) that are guaranteed not to be tracker creds.
2. **Indexer match** — for each surviving bundle:
   - Lookup tracker name in `indexer_map_override` first (highest precedence).
   - If unmatched, fuzzy-match using `Levenshtein.ratio()` (threshold ≥ 0.85, case-insensitive) against Jackett's `/api/v2.0/indexers/all/results` catalog.
   - Ambiguous matches (≥2 indexers ≥ threshold) are skipped with a clear log + entry in `skipped_ambiguous`.
3. **Configure** — for each matched indexer:
   - GET `/api/v2.0/indexers` (already-configured list). If indexer ID is present, add to `already_present`, skip.
   - GET `/api/v2.0/indexers/{id}/config` to fetch the field template Jackett expects.

- Map our credential fields onto the template (e.g., `username` → `username`, `password` → `password`, `cookieheader` → `COOKIES` value).
- POST `/api/v2.0/indexers/{id}/config` with the populated template. On 4xx, log + add to `errors`. On 5xx, one retry with 2s backoff via existing `tenacity`.

### 3.5 No new dependencies

`Levenshtein`, `aiohttp`, `pydantic`, `tenacity` are already in `download-proxy/requirements.txt`.

## 4. Data Flow

```
container boot
    └─> start-proxy.sh waits for JACKETT_API_KEY in env
        └─> python -m main
            ├─> startup hook: autoconfigure_jackett(...)
            │     ├─ scan os.environ                          [pass
1]
            │     ├─ GET /api/v2.0/indexers/all/results
(catalog)
            │     ├─ fuzzy-match bundles → indexer IDs     [pass
2]
            │     ├─ GET /api/v2.0/indexers
(already-configured)
            │     ├─ GET /api/v2.0/indexers/{id}/config
(template, per indexer)
            │     ├─ POST /api/v2.0/indexers/{id}/config   (apply,
per indexer)
            │     └─ build AutoconfigResult, log summary,
            │        store in module-level cache
            ├─> FastAPI app starts, exposes:
            │     GET /merge/jackett/autoconfig/last
(cached, redacted)
            └─> proxy + merge service serve traffic normally
```

**Persistence:** Jackett owns the durable state on disk (`./config/jackett/Jackett/Indexers/*.json`). Auto-config issues API calls; restart-survival is Jackett's job; the idempotency check on next boot prevents double-config.

**Result caching:** in-process, module-level. Lost on restart; re-run on next boot. No new files, no DB.

**Logging:** structured (existing project logger). One INFO summary per run (`autoconfigured 3/4 discovered indexers`). Per-indexer DEBUG. WARNING on errors. **No credential value at any level, ever.**

# 5. Test Matrix (CONSTITUTION 7-layer floor)

| # | Layer | File | Real infra? |
|---|---|---|---|
| 1 | **Unit** | `tests/unit/merge_service/test_jackett_autoconfig.py` | Mocks via `unittest.mock.AsyncMock`/`MagicMock` (matches existin convention in `tests/unit/test_private_tracker_searc` allowed in unit only) |
| 2 | **Integration** | `tests/integration/test_jackett_autoconfig_real.py` | Real Jackett container via `pytest-docker` |
| 3 | **E2E** | `tests/e2e/test_jackett_autoconfig_e2e.py` | Full stack (`podman compose` |
| 4 | **Security/ penetration** | `tests/security/test_jackett_autoconfig_secrets.py` + `bandit` static scan | Real stack + log files |
| 5 | **Benchmark** | `tests/benchmark/test_jackett_autoconfig_perf.py` | Real Jackett or recorded responses (allowed for benchmark stability) |
| 6 | **Automation** | `tests/contract/test_jackett_autoconfig_contract.py` (Schemathesis) | Live FastAPI |
| 7 | **Challenge** | `challenges/scripts/jackett_autoconfig_clean_slate.sh` | Full live system |

## 5.1 Layer-by-layer assertions

**Unit (1):** - Env-scan regex correctness across edge cases (lowercase ignored, mixed case, multi-underscore tracker names). - Fuzzy threshold behavior — at, above, and below 0.85. - `JACKETT_INDEXER_MAP` override precedence over fuzzy match. - Idempotency check skips already-configured indexers. - `AutoconfigResult.__repr__()` and `.model_dump_json()` redact passwords/cookies. - Errors are caught at function boundary — `autoconfigure_jackett()` never raises.

**Integration (2):** - Spin up a real Jackett container (`pytest-docker` fixture). - Inject test env vars (`TESTRACKER_USERNAME=u`, `_PASSWORD=p`). - Call `autoconfigure_jackett()`, assert it discovered the env, fuzzy-matched a public indexer Jackett ships with, configured it. - Second call: indexer in `already_present`. - `/merge/jackett/autoconfig/last` returns the live result.

**E2E (3):** - Tear down stack, wipe `./config/jackett`, set fake creds in env, `podman compose up`, wait for healthchecks. - Poll `/merge/jackett/autoconfig/last` until populated. - Assert expected indexer set configured. - Run a search through the merge service. - Assert at least one Jackett-sourced result appears in the dedup output.

**Security (4):** - Boot stack with creds in env. Capture stdout/stderr from boot. - Force errors (invalid Jackett key, malformed env) and capture exception tracebacks. - Assert: no credential value appears in boot logs, FastAPI access logs, `/merge/jackett/autoconfig/last` response, exception tracebacks, or `./config/jackett/Jackett/log.txt`. - Run `bandit` against `merge_service/jackett_autoconfig.py`. Zero high-severity findings.

**Benchmark (5):** - `pytest-benchmark` baselines: - Env-scan over 1k env vars. - Fuzzy match against a 50-indexer catalog. - Full happy-path autoconfigure round-trip. - Baselines committed to `tests/benchmark/baselines/jackett_autoconfig.json`. - Fail if any benchmark regresses > 2× baseline.

**Automation/Contract (6):** - Schemathesis loads the OpenAPI schema for `/merge/jackett/autoconfig/last`. - Fuzzes parameters; asserts no 5xx, no schema violations. - Documented mapping: CONSTITUTION says “automation” but `tests/` has no `automation/` directory. We map to `tests/contract/` (the closest existing equivalent — Schemathesis-driven contract testing of the public API surface). This mapping is recorded here so future readers don’t search for a missing dir.

**Challenge (7):** - `challenges/scripts/jackett_autoconfig_clean_slate.sh`: 1. `podman compose down --remove-orphans` 2. `rm -rf ./config/jackett` 3. `podman compose up -d` 4. Wait for healthchecks (timeout 3 min). 5. Poll `/merge/jackett/autoconfig/last` until `ran_at` populated. 6. Assert configured indexer count ≥ 1. 7. Run a search via the merge service API. 8. Assert results contain at least one Jackett-sourced entry. 9. Exit 0 on success, non-zero with diagnostics on failure.

## 5.2 Resource caps

CLAUDE.md mandatory #9 caps tests at 30-40% host. All test invocations wrapped:

```
nice -n 19 ionice -c 3 -p 1 python3 -m pytest ...
```

Container `mem_limit` already enforces in-container caps. `pytest -p no:randomly --maxfail=1 -x` for deterministic per-layer runs.

# 6. Error Handling

| Failure | Behavior | Visibility |
|---|---|---|
| Jackett unreachable at startup | Log WARNING, `errors=["jackett_unreachable"]`, **boot continues** | `/merge/jackett/autoconfig/last` shows the error |
| Jackett 401 (bad/missing key) | Log WARNING `jackett_auth_failed`, no retries | Same |
| Catalog response malformed | Log WARNING `catalog_parse_failed`, boot continues | Same |
| Credential triple incomplete (`USERNAME` only) | Skip silently at DEBUG — env may have unrelated vars | DEBUG only |
| Fuzzy match ambiguous | Skip with WARNING listing candidates, suggest `JACKETT_INDEXER_MAP` | `skipped_ambiguous` entry |
| Fuzzy match below threshold | Skip with INFO | `skipped_no_match` entry |
| `JACKETT_INDEXER_MAP` references unknown indexer ID | Log WARNING, skip mapping | `errors` entry |
| Indexer config POST 4xx | Log status + indexer ID (no body) | `errors` entry |
| Indexer config POST 5xx | One retry (2s backoff via `tenacity`), then give up | `errors` entry |
| Indexer already configured | Skip silently | `already_present` |
| Network timeout mid-call | Treated as 5xx for that one indexer; others continue | Per-indexer error |

**Universal invariants:** - `autoconfigure_jackett()` never raises out of its boundary. - Total runtime hard-capped at 60s via outer `asyncio.wait_for`. - Credentials never appear in logs, error messages, the read endpoint, or exception text.

# 7. Parity Audit Deliverable

**File:** `docs/migration/PARITY_GAPS.md`

**Method:** side-by-side read of `download-proxy/src/{api,merge_service,ui,config}` vs `qBitTorrent-go/internal/{api,service,client,models,middleware,config}`. One row per Python public function / endpoint / behavior. No code changes. Estimated 1-2 hours.

**Document shape (excerpt):**

```
# Python → Go Parity Gaps

Last audited: 2026-04-26 (commit <sha>)
Audit method: side-by-side read of public API surface; no behavior
        testing.

## Summary
- N Python features audited
- X fully ported
- Y partial
- Z missing

## Matrix
| Python module | Feature | Go location | Status
        | Risk if Go-only today |
|---------------|---------|-------------|--------|----------------------|
| `merge_service/search.py` | `start_search()` orchestrator |
        `internal/service/merge_search.go` | Ported | low |
| `merge_service/enricher.py` | TMDB title resolution | (none) |
        Missing | results lack posters/years |
| `merge_service/validator.py` | BEP 48 HTTP scrape | (none) |
        Missing | no peer-count validation |
| ... | ... | ... | ... | ... |

## Per-gap follow-up specs (proposed, in priority order)
1. Port `enricher.py` to `internal/service/enricher.go`
2. Port `validator.py` (BEP 48 + BEP 15)
3. ...
```

**Status definitions** (locked): - **Ported** — feature exists in Go with equivalent behavior. - **Partial** — feature exists in Go but lacks a sub-behavior. - **Missing** — no Go counterpart.

# 8. Verification & Push Protocol

**Order of operations** (each step is a gate; failure halts):

| # | Step | Commit + push? |
|---|---|---|
| 1 | Pre-flight: clean tree, env vars present | — |
| 2 | | — |

| # | Step | Commit + push? |
|---|---|---|
| | Tear down: containers + images + `./config/jackett`. Keep `./config/qBittorrent`, `./tmp`, `/mnt/DATA` | |
| 3 | Implement auto-config module + wire into `main.py` + add endpoint | — |
| 4 | Layer 1 (Unit) green | yes |
| 5 | Boot real stack (`podman compose up -d`), wait for healthchecks | — |
| 6 | Layer 2 (Integration) green | yes |
| 7 | Layer 3 (E2E) green | yes |
| 8 | Layer 4 (Security) green | yes |
| 9 | Layer 5 (Benchmark) green, baselines committed | yes |
| 10 | Layer 6 (Automation/Contract) green | yes |
| 11 | Layer 7 (Challenge) script exits 0 | yes |
| 12 | Parity audit: read both codebases, write `PARITY_GAPS.md` | yes |

| # | Step | Commit + push? |
|---|---|---|
| 13 | Final clean-slate verification: full tear-down + boot from zero, run challenge again, capture output into commit message | yes |
| 14 | Handoff message to project owner | — |

**Push target:**
`git push origin <branch> && git push github <branch> && git push upstream <branch>` after each gate. (All three URLs identical — push is idempotent on the remote, but we do all three so refs match locally.)

**Hard stops** (require owner input): - Any layer fails after one fix attempt. - Any push rejected. - Healthcheck unhealthy after 3 minutes. - Bench regression > 2× baseline.

# 9. File Inventory

## 9.1 New files

- `download-proxy/src/merge_service/jackett_autoconfig.py` — the module
- `tests/unit/merge_service/test_jackett_autoconfig.py`
- `tests/integration/test_jackett_autoconfig_real.py`
- `tests/e2e/test_jackett_autoconfig_e2e.py`
- `tests/security/test_jackett_autoconfig_secrets.py`
- `tests/benchmark/test_jackett_autoconfig_perf.py`
- `tests/benchmark/baselines/jackett_autoconfig.json`
- `tests/contract/test_jackett_autoconfig_contract.py`
- `challenges/scripts/jackett_autoconfig_clean_slate.sh` (executable)
- `challenges/scripts/run_all_challenges.sh` (currently no `challenges/` dir)
- `docs/migration/PARITY_GAPS.md`
- `docs/superpowers/specs/2026-04-26-jackett-autoconfig-clean-rebuild-design.md` (this file)

### 9.2 Modified files

- `download-proxy/src/main.py` — call `autoconfigure_jackett()` after Jackett key extraction, before uvicorn
- `download-proxy/src/api/routes.py` — add `GET /merge/jackett/autoconfig/last`
- `download-proxy/src/api/__init__.py` — register the new route if needed
- `docs/JACKETT_INTEGRATION.md` — append "Auto-Configuration" section
- `CLAUDE.md` — add `JACKETT_INDEXER_MAP` and `JACKETT_AUTOCONFIG_EXCLUDE` to the env vars line; note the auto-config endpoint
- `AGENTS.md` — same env additions
- `.env.example` (if exists) — add `JACKETT_INDEXER_MAP=` and `JACKETT_AUTOCONFIG_EXCLUDE=` placeholders

### 9.3 Untouched

- `qBitTorrent-go/**` (parity audit reads but does not modify)
- `frontend/**`
- `plugins/**`
- `webui-bridge.py`
- `start-proxy.sh` (unless implementation discovers a timing tweak is needed)

### 9.4 Bugfix log

Per CLAUDE.md mandatory #10 + CONST-032: any bug discovered during this work goes into `docs/issues/fixed/BUGFIXES.md` with a corresponding `challenges/scripts/jackett_autoconfig_<bug-slug>.sh` regression guard.

## 10. Out of Scope (explicit)

- Porting any Python feature to Go. The parity doc *catalogs* the gap; closing it is future work in dedicated specs.
- Go-profile rebuild or Go-side Jackett auto-config.
- Wiping `./config/qBittorrent/` or `/mnt/DATA`.
- Manual testing scripts — handoff is verbal.
- Auto-enabling Jackett's public indexer set without credentials. The Q1 decision was credentialed-only.

# 11. Open Questions for Implementation Plan

(These do not block design approval; they get resolved when `writing-plans` produces the step-by-step plan.)

- Exact branch name for this work (suggest `jackett-autoconfig-clean-rebuild`).
- Whether the existing test stack already auto-starts Jackett via `pytest-docker`, or whether the integration layer needs a new fixture.
- Whether `docs/issues/fixed/BUGFIXES.md` exists — if not, it gets created on first bugfix.
- Whether `start-proxy.sh` injects `JACKETT_API_KEY` to the proxy container before `python -m main` runs (it should, but verify at implementation time).